

Andres Mercado

BACKEND SOFTWARE ENGINEER

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Backend Software Engineer specializing in scalable APIs, async pipelines, OCR systems, and stateless architectures. At Stacksync (YC W24), delivered production-ready API integrations using async job workflows, webhooks, and Google Cloud Run. Currently building **Propia**, a multi-service backend platform for landlords that uses GPU OCR (PaddleOCR), FastAPI, async PostgreSQL, S3 storage, embeddings, and section-aware document chunking to generate structured lease data. Focused on building resilient, real-world backend systems with strong engineering fundamentals.

LANGUAGES: Python, JavaScript, SQL

FRAMEWORKS: Flask (with WTForms), Express, FastAPI

Cloud & DevOps: AWS (EC2, S3), Google Cloud Run, Docker, Nginx

DATABASES: PostgreSQL, SQLite, MongoDB

TOOLS: GitHub, Postman, SQLAlchemy

WORK EXPERIENCE

Propia >> *Founder & Backend Engineer (Personal Project)* > Remote **October 2025 – Current**

Python, FastAPI, Async SQLAlchemy, PostgreSQL, Docker, PaddleOCR, AWS S3

- **Built an async FastAPI backend** with PostgreSQL using async SQLAlchemy and Alembic migrations.
- **Architected a full async job workflow** system (queue → GPU OCR worker → DB → client sync).
- **Designed a GPU-enabled OCR pipeline** using PaddleOCR in a dedicated Docker worker.
- **Improved lease section chunking** reliability by replacing generic OCR splitting with state-specific and lease-type profiles, enabling accurate embeddings and semantic search.
- **Integrated AWS S3** for document storage and multi-step ingestion flows.
- **Orchestrated the system using Docker** Compose across API, worker, database, and GPU services.

Stacksync >> *Backend Engineer (Contract)* > Remote

June 2025 – September 2025

Python, Flask, Google Cloud Run, Webhooks

- **Delivered a production-ready API integration**, connecting scraping and extraction APIs into Stacksync's platform.
- **Engineered** multiple additional API integrations on Google Cloud Run, supporting varied authentication methods and schema-driven dynamic inputs.
- **Applied prompt engineering** techniques to structure AI-driven extraction and search responses, improving output consistency and relevance.
- **Refactored** backend logic into job ID-based workflows, enabling stateless request chaining and webhook delivery.
- **Deployed** integrations on Google Cloud Run to validate performance under real-world usage.
- **Collaborated** with GTM and product teams to align API integration workflows with customer use cases, ensuring seamless adoption.

Rabbit Aviation >> *Backend Engineer (Project: AeroSkyOps)* > Remote

April 2022 – May 2025 [GitHub](#)

Independently developed AeroSkyOps with the owner's permission, a platform designed for use at a private airport.

Flask, SQLite, PostgreSQL, AWS, Docker

- **Replaced an unstructured MongoDB** setup with PostgreSQL, introducing relational integrity for aircraft assignments, reducing manual errors, and improving database query efficiency.
- **Developed RESTful APIs** with Flask and SQLAlchemy to integrate near-live flight tracking and real-time weather updates via the FlightAware API, providing up-to-date aviation insights based on user-input tail numbers.
- **Designed a database-driven** aircraft parking assignment system that replaced manual tracking in internal testing, dynamically assigning aircraft to parking spots and improving operational efficiency.
- **Deployed Dockerized backend** on AWS EC2 with Nginx and TLS for secure API access.

Real Estate Investor >> *Owner* > Stockton, CA

October 2018 – Current

- **Achieved a six-figure income within** five years by strategically investing in and managing real estate, demonstrating strong market analysis and decision-making skills.
- **Managing a portfolio of properties** with 14 tenants, overseeing property maintenance, tenant relations, and financial management to ensure high occupancy rates and tenant satisfaction.
- **Built and led a reliable team**, coordinating various aspects of property management, from renovations to tenant communications, showcasing leadership and collaboration abilities.

EDUCATION

APP ACADEMY >> Software Engineering

Completed June 2024

Coding Dojo >> Python, MERN Stack

Completed April 2022